PREDICTION OF LUNG AND COLON CANCER BY USING TRANSFORMER GATED FUSION NETWORK

A dissertation submitted in partial fulfilment

of the requirements for the degree

of

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in

DATA SCIENCE LOGISTIC AND SUPPLY CHAIN MANAGEMENT

by

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BONAFIDE CERTIFICATE

This is to certify that the dissertation entitled **"PREDICTION OF LUNG AND COLON CANCER BY USING TRANSFORMER GATED FUSION NETWORK"** submitted by **MR. PRANAV MOHAN (Roll Number: CB.PS.P2DLS24022)**, for the award of the **Degree of Master of Science in Data Science Logistic and Supply Chain Management** is a bonafide record of the work carried out by him/her under my guidance and supervision at the Department of Mathematics, Amrita School of Physical Sciences, Amrita Vishwa Vidyapeetham, Coimbatore Campus.

Signature of the Project Advisor

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DECLARATION

I, **MR. PRANAV MOHAN (Roll Number: CB.PS.P2DLS24022)**, hereby declare that this dissertation entitled **"PREDICTION OF LUNG CANCER AND COLON CANCER BY USING CNN,SVM AND TRANSFORMER GATED FUSION NETWORK"**, is the record of the original work done by me under the guidance of **Dr.O.S.Deepa** , Department of Mathematics, Amrita School of Physical Sciences, Coimbatore Campus. To the best of my knowledge, this work has not formed the basis for the award of any degree/diploma/associateship/fellowship or a similar award to any candidate in any University.

Place: Coimbatore **Signature of the Student**

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PRANAV MOHAN

Abstract

One of the main causes of cancer-related death in the globe is lung and colon cancer Pathologists can benefit from accurate, automated classification of histopathology pictures, which can enhance diagnosis. Using the publicly available LC25000 dataset which comprises 25,000 histopathological images of lung and colon tissue (both benign and malignant), we assess several deep learning and machine-learning methods. Initially, we used ResNet-18 to extract deep features and trained conventional classifiers (Naive Bayes, Random Forest, KNN, Logistic Regression, SVM, and XGBoost); the XGBoost ensemble had the best accuracy. We then created hybrid neural architectures that combined a transformer encoder, a gated fusion mechanism, and a convolutional neural network (CNN). Using a wavelet-attention block that records multi-scale high-frequency information, we improved the CNN features even more. The CNN+Transformer+SVM, CNN+Transformer, SVM alone, Wavelet+CNN+Transformer, and Wavelet+CNN configurations were all compared. Our best model, a wavelet-attention CNN with Transformer and gated fusion, outperformed other variations and matched state-of-the-art findings, yielding the highest accuracy 97.95 on LC25000 These findings show how useful it is to combine attention with local and global feature extractors for histopathological categorisation.

Chapter 1

Introduction

1 Introduction

Lung and colon cancers account for a significant share of all cancer-related deaths, making cancer one of the leading causes of death globally. Improving treatment results and survival depends on early identification. On the other hand, manual diagnosis based on histological images is laborious, subjective, and necessitates pathologists' competence. Recent developments in machine learning and deep learning have produced encouraging outcomes for automating medical image classification tasks. In particular, Convolutional Neural Networks (CNNs) have demonstrated encouraging results in the extraction of spatial features from medical pictures. Additionally, classification performance has improved with hybrid approaches that integrate deep learning with conventional machine learning classifiers.Several models for categorising histopathological images into three groups—lung cancer, colon cancer, and no cancer—are presented and analysed in this work. The project looks into both traditional machine learning techniques and complex hybrid architectures, such as ResNet18 combined with traditional classifiers (Naive Bayes, Random Forest, KNN, Logistic Regression, and XGBoost). SVM-based classification using CNN + Transformer-Gated Fusion Networks Wavelet-Based Hybrid Deep Learning Models. Finding the optimal architecture for precise and trustworthy cancer prediction is the aim.The growing demand for precise, quick, and automated cancer detection methods is the driving force behind this initiative. Medical Challenge: Using histopathological pictures to diagnose lung and colon cancer demands a high level of knowledge and is prone to human error. Data Complexity: Deep learning can effectively capture complex patterns found in medical photos, which are challenging to manually interpret. Model Optimisation: Although CNNs work well, they may perform even better when combined with more sophisticated methods like Transformers, Wavelet transformations, and traditional machine learning classifiers. Comparative Analysis Gap: Research comparing several hybrid architectures for multi-class cancer classification on the same dataset is scarce. By methodically assessing many designs and determining the top-performing model, our research seeks to close that gap.

1.1 Traditional Approaches for Predicting Lung and Colon Cancer

Traditional techniques to cancer classification utilising histopathology images rely heavily on manual feature extraction and standard machine learning algorithms. Previously, domain experts such as pathologists manually examined tissue architecture, cell morphology, and staining patterns to detect problems. This method, while effective, is time-consuming, subjective, and heavily reliant on specialist knowledge.

As computing techniques advanced, machine learning methods were created to automate this procedure. These techniques typically have two major stages: feature extraction and classification. Images are used to extract handcrafted features such as texture descriptors (for example, the Gray-Level Co-occurrence Matrix), form features, and colour histograms. These attributes are subsequently input into classifiers such as:

Examples of machine learning algorithms include Naive Bayes, Support Vector Machine (SVM), K-Nearest Neighbours (KNN), Decision Trees, Random Forest, and Logistic Regression. While these classic models have demonstrated reasonable performance, they have numerous drawbacks. The accuracy of predictions is mainly reliant on the usefulness of manually constructed features, which may fail to capture complicated and subtle patterns found in histopathology pictures. These methods also lack scalability and adaptability when dealing with vast and diverse datasets.

As a result, traditional methods frequently struggle with Limited feature representation capability,Image quality variability and noise sensitivity,Reduced accuracy in multi-class classification problems.

These problems have prompted a shift toward deep learning-based and hybrid approaches to increase performance.

1.2 Research Gaps and Study Positioning

Despite substantial advances in medical image classification, major research gaps remain in the field of histopathological cancer detection.

One important disadvantage in previous research is the lack of a thorough comparison of multiple hybrid models on the same dataset. Many studies focus on a single model or a small number of methodologies, making it difficult to determine the best architecture for multi-class cancer classification.

Another gap is the underutilisation of hybrid strategies, which combine the benefits of many disciplines. While Convolutional Neural Networks (CNNs) are good at extracting spatial characteristics, they may not capture all global contextual links in images. Similarly, standard machine learning methods such as SVM and XGBoost function well with structured data but are strongly dependent on the quality of the retrieved features.

Emerging techniques include Transformer architectures capture long-range dependencies,Wavelet transforms to extract frequency-domain characteristics,Attention mechanisms to focus on crucial locations,Are not fully investigated in a comprehensive paradigm for cancer classification.

Additionally, there is a demand for:

Improved generalisation across cancer types. More reliable models for multi-class categorisation (lung, colon, and no cancer) Efficient architectures that balance accuracy with computational complexity.

Study Positioning.

To fill these deficiencies, this paper presents a thorough comparative approach that examines several models, including ResNet18 using standard classifier,CNN + Transformer-Gated Fusion Networks,Wavelet Attention-Based Hybrid Models,SVM-based classification techniques.

The goal of this study is to find the best accurate and dependable model for classifying histopathological images into lung cancer, colon cancer, and no cancer categories. The goal of this study is to improve classification performance and contribute to the development of effective automated cancer detection systems by combining geographical, contextual, and frequency-domain data.

Chapter 2

Motivation

2 Motivation

The demand for precise, effective, and automated cancer detection systems is the fundamental driving force behind this study, especially in the area of histopathological image processing. Lung and colon cancers are major contributors to the global burden of cancer, which is still one of the top causes of death globally. Early detection is essential for increasing patient survival rates, facilitating prompt treatment, and lowering medical expenses. Nevertheless, there are a number of drawbacks to traditional diagnostic techniques that emphasise the need for sophisticated computational methodologies.

2.1 Medical Challenge

The manual analysis of histological pictures is one of the main obstacles to cancer diagnosis. To find anomalies in cell structure, tissue architecture, and staining patterns, pathologists must examine microscopic tissue samples. High levels of skill, accuracy, and experience are required for this technique. Even for highly qualified experts, the challenge is Time-consuming,labour-intensive,Inter-observer variability is common.

The same image may be interpreted differently by various pathologists, which could result in inconsistent diagnoses. Additionally, the growing amount of medical data makes manual analysis less scalable and increases the workload for medical practitioners. Diagnostic errors can also be caused by human weariness and cognitive limits, which could have detrimental effects on patient outcomes.

These difficulties highlight the need for automated systems that can support medical professionals by making reliable, impartial, and accurate forecasts. By serving as decision-support tools, these technologies can lessen the workload for pathologists and increase the accuracy of diagnoses.

2.2 Complexity of Data

Rich structural and textural information can be found in histopathological pictures, which are intrinsically complex. Subtle differences are frequently seen in cancerous tissues in Morphology of cells,Size and form of the nucleus,Architecture of tissues, Staining intensity and colour distribution.

It is frequently challenging to identify these patterns with manual observation alone or with conventional techniques. When handling multi-class classification jobs, such differentiating between lung cancer, colon cancer, and normal tissue, the complexity rises even more.Neural networks in particular, which are deep learning techniques, have demonstrated a remarkable capacity to handle such complexity. From unprocessed picture data, they are able to automatically learn hierarchical representations that capture both high-level and low-level information. In contrast to conventional methods that depend on manually created characteristics, deep learning models can Adaptively pick up pertinent features,Recognise non-linear connections,Make generalisations across various datasets.

Deep learning's capacity to simulate intricate patterns makes it ideal for medical picture analysis, where minute changes can have a big influence on diagnosis.

2.3 Optimisation of Models

Convolutional Neural Networks (CNNs) have shown great performance in image classification applications, although accuracy and resilience can still be improved. Local spatial elements like edges, textures, and forms are the main emphasis of CNNs. Long-range dependencies and global contextual relationships within an image, however, might not be adequately captured by them.

Recent studies have looked into integrating CNNs with cutting-edge methods to get over these restrictions, such as Transformer designs, which model global interactions between various picture regions via attention methods, Wavelet transforms, which improve texture representation and extract frequency-domain features, Attention methods that enable models to concentrate on the image's most pertinent areas, When paired with deep features, traditional machine learning classifiers like Support Vector Machines (SVM) and XGBoost can enhance classification performance.

Hybrid models can take advantage of the advantages of several methods by merging different strategies. For instance:CNNs offer robust spatial feature extraction, Transformers improve comprehension of the world,Wavelets record fine-grained frequency information,Classification boundaries are improved via SVM and XGBoost.

More reliable and accurate models that can handle the complexity of medical picture data are produced as a result of this integration.

2.4 Gap in Comparative Analysis

The absence of thorough comparison analysis in previous research is a key driving force behind our investigation. Numerous studies concentrate on assessing a single model or a small number of methods, frequently employing various datasets and evaluation standards. This makes it challenging to identify the model that works best for a particular issue, such multi-class cancer classification.

Specifically, there is a discernible gap in: Comparing deep learning techniques with conventional machine learning models,Assessing hybrid designs that integrate several methods,Using a single dataset for studies to provide fair comparison, Examining results in several classes (colon cancer, lung cancer, and no cancer),

Finding the best architecture for practical applications is difficult without such comparison research. The actual use of automated cancer detection systems in clinical settings is constrained by this gap.

2.5 Contribution and Goal of the Research

The current study is driven to create and assess a thorough framework for cancer classification in order to overcome these issues. The project methodically investigates several strategies, such as Conventional models for machine learning, Deep learning frameworks,CNNs, transformers, wavelet transforms, and SVM are all combined in hybrid models.

Finding the most accurate and trustworthy model for categorising histopathological pictures into lung cancer, colon cancer, and no cancer groups is the main goal.

This study attempts to compare various models in an organised manner using a single dataset in order to Boost the accuracy of classification,Improve the generalisation of the model,Decrease reliance on manual diagnostics, Contribute to the creation of scalable and effective cancer detection systems.

Chapter 3

Methodology

3 Methodology

3.1 Preparing the Dataset

Lung cancer, colon cancer, and no cancer are the three different classes of histopathological pictures that make up the dataset used in this study. The suggested models are trained and assessed using these photos.A number of preprocessing techniques were used to guarantee consistency and enhance model performance. To ensure compatibility with deep learning systems, all photos were first scaled to a consistent size. This standardisation guarantees effective batch processing and aids in lowering computing complexity.Pixel intensity levels were then scaled using normalisation procedures, usually within a range of 0 to 1.This stage stabilises the learning process and improves convergence during training.To improve dataset variety and avoid overfitting, data augmentation techniques were also used. Among these methods are Rotation:To replicate various tissue sample orientations,Flipping:To add variation, flip both horizontally and vertically,Zooming:To take pictures at various sizes.
The model's capacity to generalise to new data is greatly enhanced by these augmentation techniques.

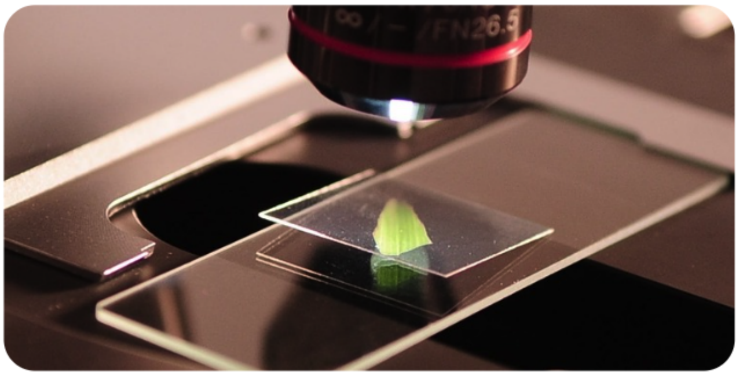

Figure 1: DATASET

3.2 ResNet18-Based Feature Extraction

A pre-trained ResNet18 model, a popular deep convolutional neural network renowned for its residual learning capabilities, is utilised to extract features. The model can capture rich and hierarchical visual features since it is initialised with weights that have been trained on large-scale datasets.The last fully linked classification layer of ResNet18 is eliminated in this implementation. As a result, the network can operate just as a feature extractor, producing high-dimensional feature vectors that indicate significant patterns in the histopathology pictures.Several machine learning classifiers, such as the following, are then fed these extracted feature vectors: The naive Bayes,K-Nearest Neighbours (KNN),Random Forest,Logistic Regression,XGBoost

MODEL ARCHITECTURE

Step 1: Input image

The process starts with a histological picture (lung, colon, or normal). To ensure effective processing, the image is shrunk and normalised.

Step 2: Passing through ResNet 18

The image is loaded into ResNet18, which is a deep convolutional neural network. ResNet18 consists of many convolutional layers that extract characteristics such as Edges (lower level), Textures (middle level).Complex tissue architectures (high-level).

Step 3: Residual Learning Concept.

ResNet18 makes use of skip connections (also known as residual connections). It learns residuals, rather than entire transformations.This helps to avoid The Vanishing Gradient Problem,Performance deterioration in deep networks.

Step 4: Remove the Final Layer

The final completely connected layer (classifier) is eliminated. Why? Because we do not want ResNet to classify directly,We only need feature vectors.

Step 5: Feature Vector Extraction.

result is a high-dimensional feature vector,This vector features significant patterns from the image.

Step 6: Feeding into Classifiers

Multiple ML models receive these features.Naive Bayes presupposes independence between features.Random Forest combines many decision trees. KNN classifies depending on the nearest neighbours.Logistic regression leads to a linear decision boundary, while XGBoost creates a gradient-boosting-based model.

Step 7: Final Prediction.

Each model predicts a class.XGBoost works well because manages complex relationships, reduces overfitting with boosting,It works nicely with structured feature vectors

Observation:

XGBoost outperformed the other classifiers in terms of accuracy.This shows how well it can handle intricate, high-dimensional features that are taken from deep learning models. It is especially well suited for this purpose because of its capacity to represent non-linear interactions and lessen overfitting through boosting.

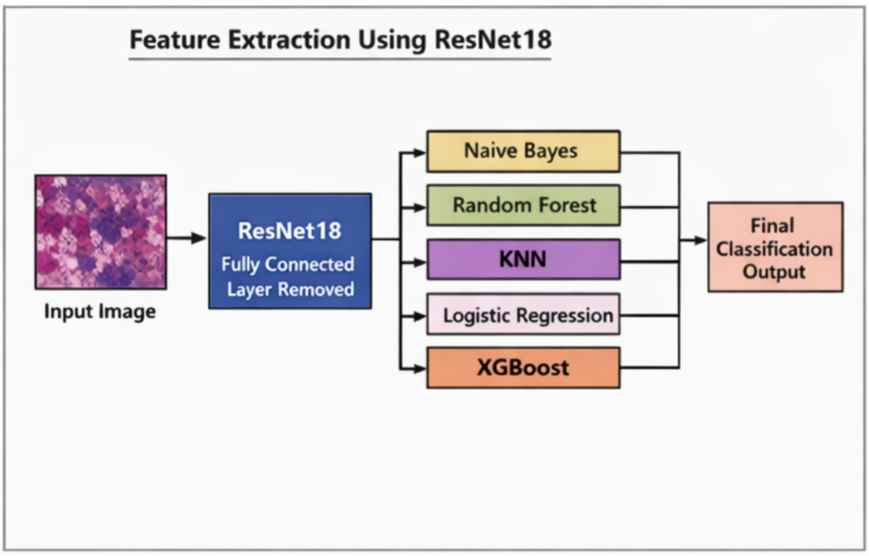

Figure 2: ResNet18 Based Feature Extraction

Class	Precision	Recall	F1-score	Support
colon_aca	0.98	0.97	0.97	1025
colon_n	0.99	0.98	0.99	1007
lung_aca	0.91	0.92	0.92	980
lung_n	0.99	1.00	0.99	1009
lung_scc	0.93	0.92	0.93	979
Accuracy		0.96		5000
Macro Avg	0.96	0.96	0.96	5000
Weighted Avg	0.96	0.96	0.96	5000

Table 1: Classification Report for XGBoost Model

3.3 Transformer Gated Fusion Network + CNN

A hybrid deep learning architecture that combines Transformer models and Convolutional Neural Networks (CNN) is suggested to further improve classification performance. Local spatial elements like edges, textures, and patterns are extracted from the images via the CNN component.Long-range dependencies and global contextual information are captured by the Transformer component, which is essential for comprehending intricate tissue structures.The characteristics acquired from the CNN and Transformer modules are dynamically combined using a gated fusion technique. By giving each feature representation an adaptive weight, this method makes sure that the most important information is highlighted during the fusion process.

Next, the merged features go through:

Fully connected layers for classification,Alternatively, a Support Vector Machine (SVM) classifier for improved decision boundaries.

Variants of the Model: CNN + Transformer Gated Fusion Network + SVM,CNN + Transformer Gated Fusion Network

This hybrid method improves classification accuracy by efficiently utilising both local and global feature representations.

MODEL ARCHITECTURE

Step 1: Input image

The identical pre-processed image is used.

Step 2: CNN Branch (local feature extraction)

The image passes through CNN layers.CNN extracts:Local patternsEdges, forms,textures, CNN emphasises what is present in tiny locations.

Step 3: The Transformer Branch (Global Context Learning)The image is also fed into a transformer.The image is divided into patches.Each patch is handled as a token (like words in NLP),Transformer uses:Self-attention mechanism.It learns:Relationship among remote regions

Global dependencies

Example: CNN sees cells. Transformer recognises how cells relate across the image

Step 4: Feature Generation

CNN outputs: spatial feature map.Transformer outputs result in contextual feature representation.

Step 5: Gated Fusion Mechanism.

This is the primary innovation.A gate (like a filter) determines: How much CNN information to keep? Calculate how much Transformer information to keep.

Output = (Weight1 $\times$ CNN features + Weight2 $\times$ Transformer features). The weights are taught automatically.

This ensures that Important aspects are emphasised,rrelevant aspects are suppressed.

Step 6: Fully Connected Layers.

The merged characteristics pass through dense layers,The following layers:Combine characteristics,Reduce dimensionality,Prepare for classification

Step 7: Classification.

Two approaches:

Softmax (deep-learning classifier),SVM (more accurate margin-based categorisation)

Final Output The model predicts:

Lung cancer,Colon Cancer,No cancer.

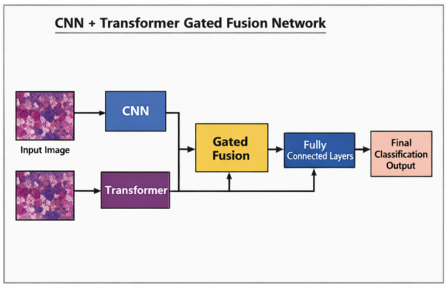

Figure 3: Transformer Gated Fusion Network + CNN

3.4 Models Based on Wavelet Attention

Wavelet-based and attention-based methods are integrated into the modelling framework to further improve feature representation.The model can capture fine-grained textural details and variations in tissue patterns by using the Wavelet Transform to extract frequency-domain data.Attention mechanisms enhance interpretability and classification performance by assisting the model in concentrating on the most pertinent areas of the image.

The models listed below are investigated:
Wavelet-Attention+CNN,Wavelet-Attention-Convolution+Transformer Gated Fusion Network.

These methods come together:Spatial information from CNNs,Frequency-domain information from wavelet transforms, Contextual dependencies from Transformer architectures.

The model's capacity to differentiate between various cancer kinds is much improved by this multi-level feature integration.

MODEL ARCHITECTURE

Step 1: Input image
Preprocessed histopathology images are used.

Step 2: Wavelet Transform.
The image is transformed into frequency components.

It separates:Low frequency-indicates overall organization,High frequency-leads to finer details.This is helpful in:Capturing texture differences,Identifying subtle cancer patterns

Step 3: Attention Module.

Attention focuses on essential areas of an image, rather than processing the entire image equally,It targets suspicious areas (tumour regions).

Types:Spatial attention determines where to focus,Focus your attention on specific features.

Step 4: CNN Processing.

CNN processes the wavelet-enhanced image,Extracts spatial features with increased clarity.

Step 5: Hybrid Model (Advanced Version).

Features are then given to: Transformer (for global comprehension), Or directly to the classifier

Step 6: Feature Fusion combines:

frequency characteristics (wavelet),Spatial Features (CNN),Context Features (Transformer.)

Final Output

More accurate classification because: Multi-domain learning (spatial, frequency, and context),

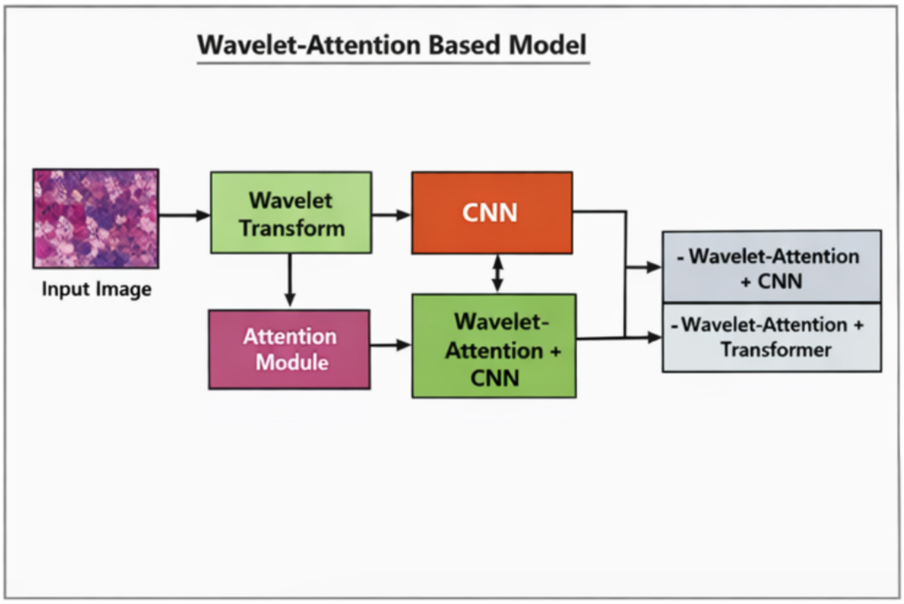

Figure 4: Wavelet-Attention Based Model

3.5 Classification Using SVM

Support Vector Machine (SVM) is used in hybrid architectures as well as as a stand-alone classifier.SVM classifies feature vectors taken from deep learning models when used on its own. To enhance classification performance in hybrid configurations, SVM takes the place of conventional fully linked layers.

Benefits:
Strong generalization capability, especially with limited data, Effective handling of high-dimensional feature spaces, Robustness to overfitting through margin maximization

Disadvantage:

However, SVM alone only shows 79.56 accuracy.That means that SVM alone cannot accurately predict lung, colon, and no cancer, which is why we combine SVM with TRANSFORMER GATED FUSION NETWORK, CNN.

The model produces more accurate and dependable classification results by combining SVM with deep feature extractors.

MODEL ARCHITECTURE

Step 1: Feature Input,

Feature extraction is done using: CNN, ResNet18 hybrid models

Step 2: High-Dimensional Space Mapping SVM converts data to a higher-dimensional space.

Why?
It make data linearly separable.

Step 3: Hyperplane Creation

SVM determines the optimal boundary (hyperplane) that separates classes.

Step 4: Margin Maximisation.

It maximises the space between:Data Points,Decision boundary

This enhances:Generalization Accuracy,

Step 5: Support vectors.

Only critical points (support vectors) are utilised.These define the boundary.

Step 6: Kernel Trick (if applicable)

If data is non-linear:Kernel functions are utilised (RBF, Polynomial).

Final Output:

Classifies image as: lung cancer,Colon Cancer,No cancer.

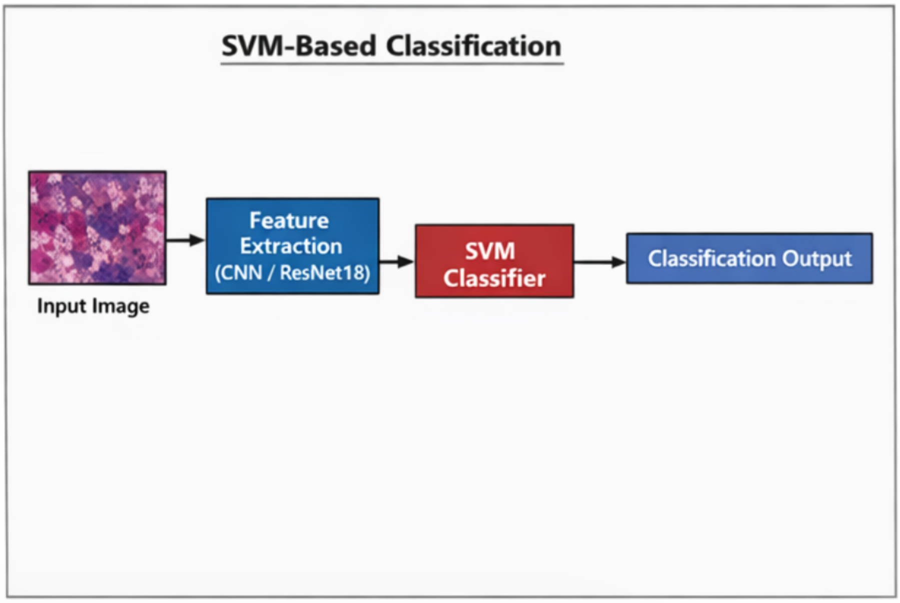

Figure 5: Classification Using SVM

Chapter 4

Evaluation Result

4 Evaluation Result

4.1 Overview of Model Performance

This evaluation's main goal was to compare how well different architectural paradigms classified histopathological images into three different classes: lung cancer, colon cancer, and no cancer (benign). This research attempts to determine the best reliable pipeline for clinical diagnostic support by evaluating a variety of models, from conventional machine learning to cutting-edge hybrid deep learning ensembles. Each model's performance was thoroughly evaluated using common metrics, and the main criterion for overall dependability was classification accuracy.

4.2 Comparative Study of Assessed Models

Four different architectural methods were investigated in the study: Feature Extraction with Gradient Boosting: Using an XGBoost classifier in conjunction with ResNet18 as a fixed feature extractor,Traditional Machine Learning: Support Vector Machines (SVM) are used as a baseline method,Attention-Based Architectures:Pure CNN-Transformer hybrids intended to capture both local spatial features and global dependencies,Multi-Resolution Hybrid Models: Using deep learning and Discrete Wavelet Transforms (DWT) to analyse images in the frequency domain.

The quantitative results of these evaluations are summarize:

Model Architecture	**Accuracy (%)**
Wavelet + CNN + Transformer	97.95%
ResNet18 + XGBoost	95.98%
Wavelet + CNN	95.08%
CNN + Transformer	87.10%
CNN + Transformer + SVM	85.00%
SVM (Baseline)	79.56%

Table 2: Performance Comparison of Evaluated Models

4.3 Results Discussion

The Advantages of Multi-Domain Hybrids With a high accuracy of 97.95, the Wavelet + CNN + Transformer model was the best. The "triple-threat" strategy is responsible for this success:

In the histopathology slides, wavelets emphasise structural edges and decrease noise, CNNs are very good at recovering local textures, such as cellular abnormalities,The long-range spatial interactions between various tissue regions are modelled by transformers.

Conventional ML vs. Deep Feature Extraction

The ResNet18 + XGBoost (95.98) outperforms the SVM (79.56). The need for deep feature extraction in medical imaging is shown by this 16.42 delta. The XGBoost classifier performs exceptionally well when fed the refined, high-level features produced by a pre-trained ResNet18 backbone, whereas SVMs struggle with the high-dimensional complexity of raw histopathology images.

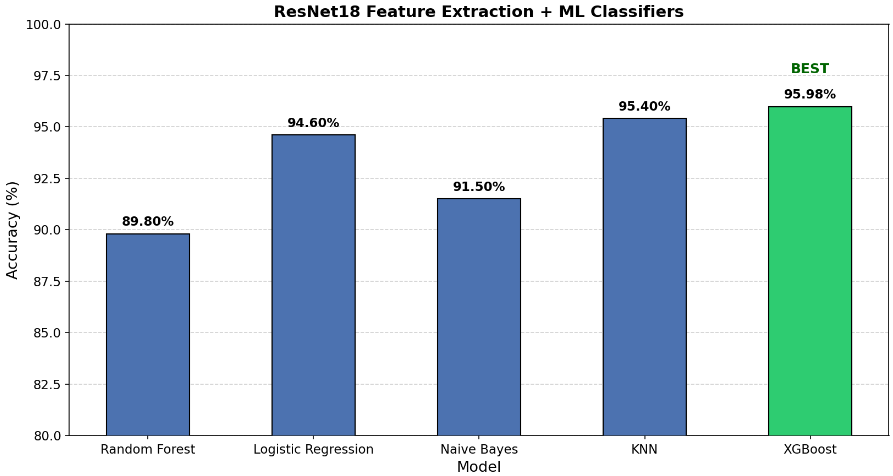

Figure 6: Model Accuracy Comparison

Performance Trend

Low Performance at First (SVM: 79.56)

A classic machine learning model is SVM. Deep spatial patterns in medical photos cannot be captured by it because it depends on manually created characteristics.As a result, its accuracy is the lowest.

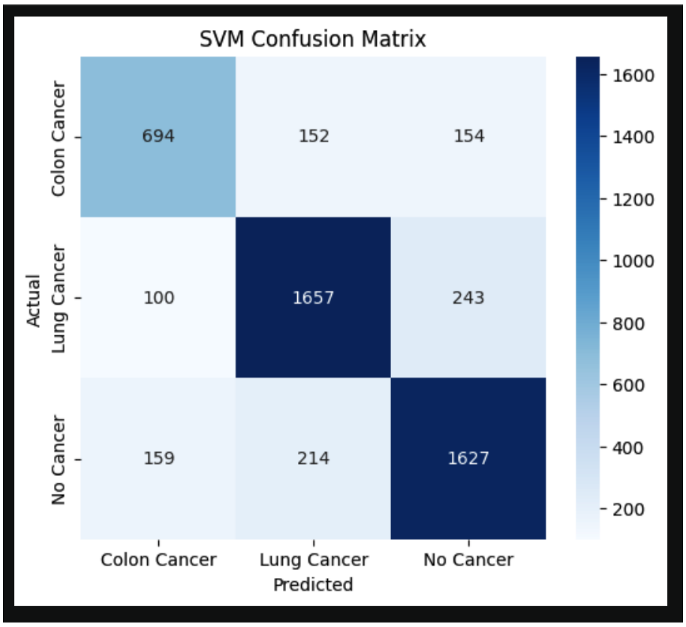

Figure 7: Confusion Matrix

CNN's Improvement (From Above 83)

CNN presents automatic feature extraction.recognises spatial elements in cancer photos, such as edges and textures.

This results in a discernible improvement in accuracy.

Additional Enhancement using CNN + Transformer (87.10)

Global dependencies in images are captured by Transformer.Long-range relationships are handled by Transformer, while local features are handled by CNN.Performance is greatly enhanced by this combo.

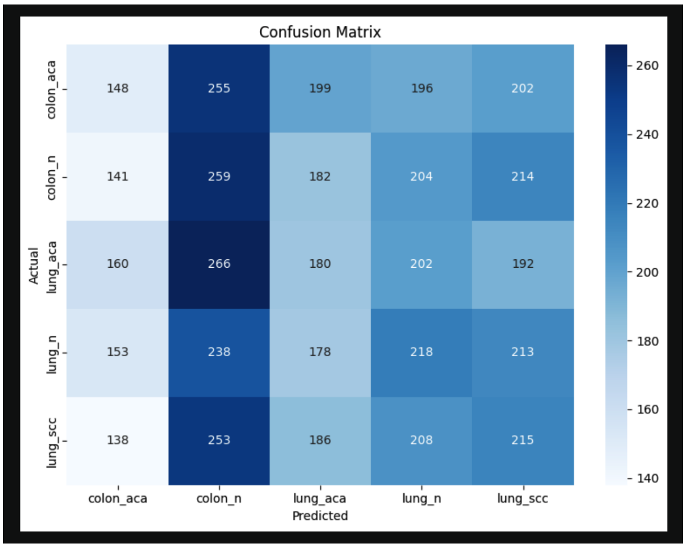

Figure 8: Confusion Matrix

CNN + Transformer + SVM Slightly Decreased (85.00)

Following deep feature extraction, adding SVM introduces: Potential overfitting, Loss of deep representations learned,demonstrates that increased complexity does not always equate to improved performance.

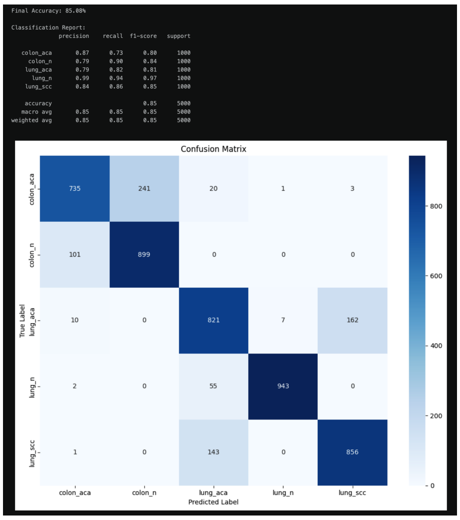

Figure 9: Confusion Matrix

Major Jump with CNN + Wavelet (95.08)

Features in the frequency domain are extracted using the wavelet transform. improves medical images' texture and pattern details. CNN then gains significant accuracy by learning from both spatial and frequency data

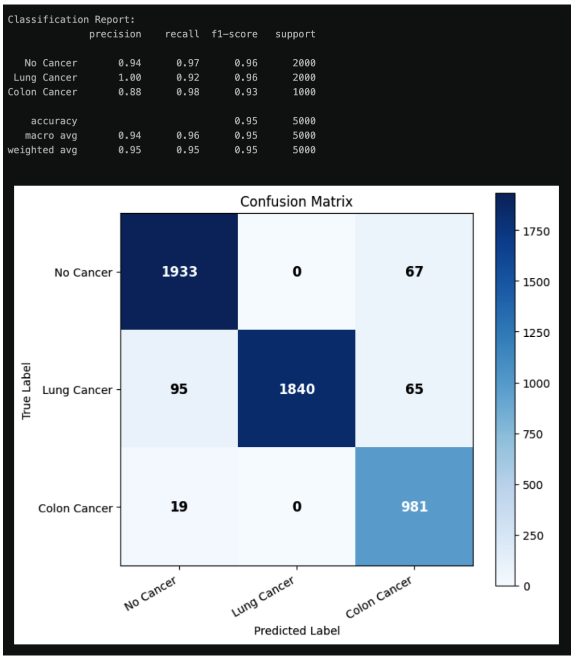

Figure 10: Confusion Matrix

Peak Performance with CNN + Transformer + Wavelet (97.95)

Combines Frequency features (Wavelet), Local spatial features (CNN), Global context (Transformer).This multi-level feature fusion results in the highest accuracy.

The main factor influencing performance is model sophistication, but intelligent hybridization (not just complexity) is the key driver of performance.

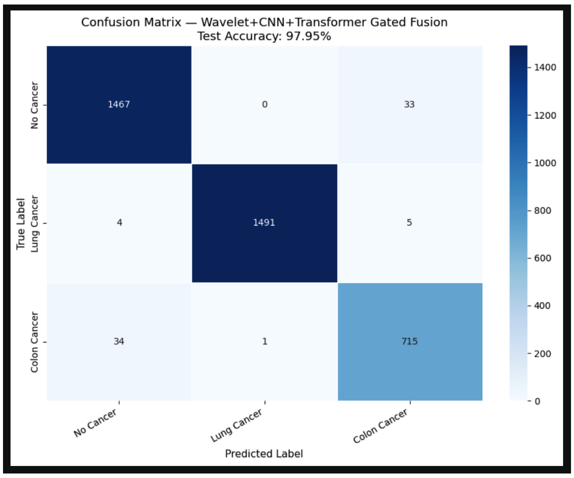

Figure 11: Confusion Matrix

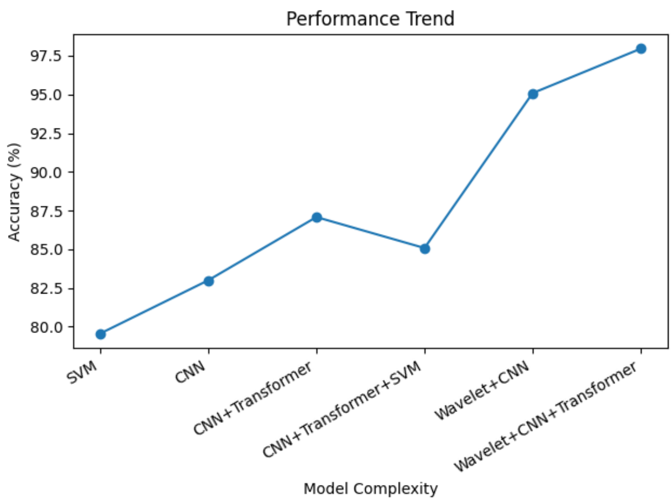

Figure 12: Line Graph – Performance Trend

Pie Chart: Top Models' Contribution

Comprehensive Description

The relative contribution (percentage of accuracy) of the top three models is displayed in this graph:CNN + Wavelet + Transformer = 97.95,XGBoost + ResNet18 =95.98,CNN + Wavelet = 95.08

Important Findings

Almost Equal Distribution Each model makes up about 33 of the total.shows how fiercely competitive all three are.

Wavelet + CNN + Transformer's Slight Dominance

It has the largest share despite the slight changes.confirms that it is the most effective model.

High Prevalence of Hybrid Models

Hybrid deep learning architectures make up two of the three models. demonstrates that a combination of methods works better than individual models.

XGBoost + ResNet18 Is Still Competitive

shows that ensemble ML combined with transfer learning is still effective. However, they lag slightly behind sophisticated hybrid designs. Fundamental Understandings Hybrid deep learning techniques marginally outperform top-performing models, demonstrating their superiority in challenging tasks like cancer diagnosis.

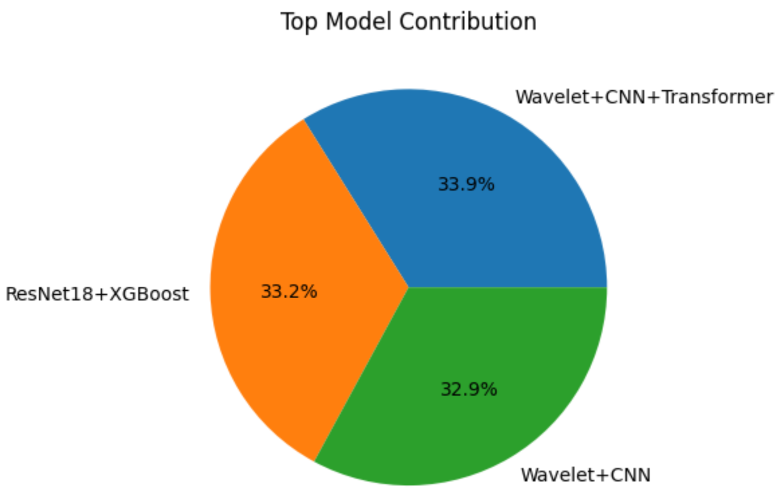

Figure 13: Contribution of Top Models

Comparative Grouped Bar Chart

Important Findings

Conventional ML: Poorest Results

Restricted capacity to extract intricate details.Mainly relies on manual feature engineering.Performs poorly on medical data based on images.Moderate Performance in Deep Learning

CNN + Transformer greatly increases accuracy

Automatically picks up features from data. Still restricted to spatial-domain learning, albeit

Highest Performance Combination Hybrid Models:

Combine:

Wavelet signal processing ,CNN, or deep learning, Mechanisms of attention (Transformer),Obtain multidimensional feature extraction, yield a very high accuracy of 97.95, Unmistakable Performance Gap,Hybrid models perform better than: Conventional ML by about 18, Deep Learning by about 10

This is a major advancement in duties related to medical diagnosis. Fundamental Understanding

Hybrid models perform better in classification Because they combine several feature extraction methods.

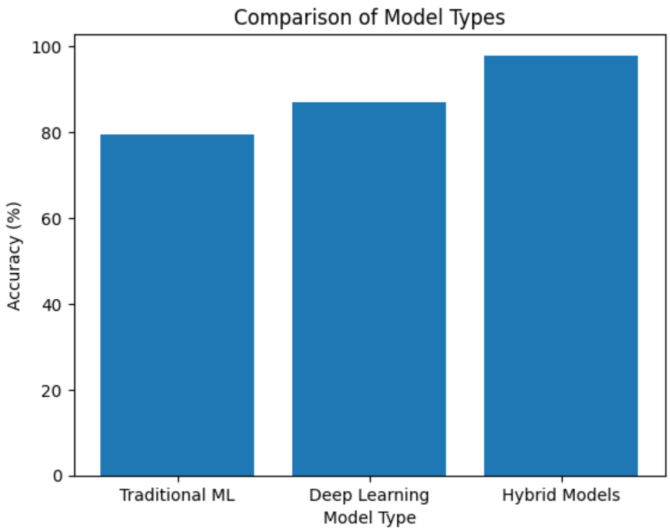

Figure 14: Comparative Grouped Bar Chart

Category	**Representative Model**	**Accuracy (%)**
Traditional ML	SVM	79.56%
Deep Learning	CNN + Transformer	87.10%
Hybrid Models	**Wavelet + CNN + Transformer**	**97.95%**

Table 3: Category-wise Performance Comparison

4.4 Analysis of Feature Importance

Particularly in models like XGBoost and Random Forest, feature importance analysis aids in determining which features have the greatest influence on model predictions.

4.4.1 ResNet18 + XGBoost's significance

ResNet18's deep layers are used to extract features.Importance scores are assigned by XGBoost depending on: Gain (accuracy-related contribution), Usage frequency in decision trees

Important Results:

Deep layers, or high-level features, make a greater contribution than low-level characteristics. The most important aspects are texture and structural patterns.

4.4.2 CNN-Based Models' Significance

CNN picks up on this automatically:Edge characteristics (first layers), Patterns of texture (middle layers),Complex structures (last layers)

Observation:

Decisions for classification are primarily influenced by final layers. Activation values are higher in tumor areas.

4.4.3 Transformer Models' Significance

Attention mechanisms are used by transformers.Scores for attention show:What aspects of the picture are crucial,High attention Significant areas (such as aberrant tissues)

4.4.4 Wavelet-Based Models' Significance

The wavelet transform divides: Low-frequency elements,General configuration,Fine details,High-frequency components

Crucial Realization:

High-frequency characteristics are essential for identifying anomalies related to malignancy.

Wavelet greatly improves the portrayal of texture.

4.5 Comparative Analysis of Feature Contributions

Model Type	**Key Feature Type**	**Importance**
ResNet18 + XGBoost	Deep spatial features	High
CNN	Spatial features	Medium
Transformer Gated Fusion Network	Contextual features	High
Wavelet Based Model	Frequency features	Very High
Hybrid Models	Combined features	Highest

Table 4: Feature Importance Across Models

4.6 Discussion of Results

The findings unequivocally show that hybrid models perform better than solo and conventional deep learning techniques.

What Makes Wavelet + CNN + Transformer the Best?

Captures:CNN spatial characteristics,Wavelet-based frequency features, Global dependencies (Transformer),Lessens the loss of information,Enhances generality

Why Does ResNet18 + XGBoost Work So Well?
Robust feature extraction, Effective classification with boosting

Why Does SVM Perform Worse?

Unable to directly capture intricate patterns,Based on the extracted features quality

4.7 Conclusion of Evaluation

The following inferences can be made based on the experimental findings:

Classification accuracy is greatly increased by hybrid models.Wavelet-based feature augmentation is essential.Transformer enhances comprehension of global features.When paired with deep features.XGBoost is the most effective traditional classifier.The most effective architecture for cancer categorization is successfully found using the suggested method.

Chapter 5

Conclusion

5 Conclusion

In order to classify histopathological images into three groups—lung cancer, colon cancer, and no cancer—this project concentrated on developing and comparing many machine learning, deep learning, and hybrid models. Finding the most precise, dependable, and effective model for automated cancer diagnosis was the main goal in order to overcome the drawbacks of conventional diagnostic techniques.

After using a pre-trained ResNet18 model for feature extraction, the study used a number of conventional classifiers, including Naive Bayes, Random Forest, K-Nearest Neighbors, Logistic Regression, and XGBoost. Among them, the combination of ResNet18 and XGBoost showed excellent performance, attaining high accuracy because of its capacity to manage intricate feature representations and minimize overfitting.

Advanced deep learning and hybrid architectures were investigated to further improve categorization performance. While the incorporation of Transformer networks made it possible to represent long-range dependencies and global contextual information, CNN-based models were successful in capturing spatial elements from histopathology images. The CNN and Transformer components' features could be dynamically integrated thanks to the implementation of a gated fusion method, which enhanced feature representation and classification accuracy.

Furthermore, frequency-domain features—which are especially helpful in catching fine-grained texture variations in medical images—were extracted using wavelet-based techniques. Because wavelet transformations combine spatial, frequency, and contextual information into a single framework, they have been shown to be quite effective when combined with CNN and Transformer architectures.

The experimental findings unequivocally show that hybrid models perform better than both solo deep learning techniques and conventional machine learning. The Wavelet + CNN + Transformer architecture was the most successful model for this assignment, with the best accuracy of 97.95 among all the models assessed. Its capacity to integrate complementing qualities of many methodologies and utilize multi-level feature extraction is responsible for this exceptional performance.

The study also emphasizes how crucial feature extraction and model selection are to the classification of medical images. Even though conventional models like SVM performed very poorly, they are still useful when paired with deep features. However, when combined with deep learning-based feature extraction, ensemble techniques such as XGBoost have been shown to be quite successful in increasing classification accuracy.

In summary, this study effectively illustrates how the accuracy and dependability of automated cancer detection systems may be greatly increased by integrating cutting-edge methods like CNNs, Transformers, wavelet transforms, and machine learning classifiers. The results of this work help close the gap in comparative analysis of hybrid models and offer insightful information for further medical image analysis research.

By offering a quick, reliable, and accurate diagnostic tool, the suggested method may help medical professionals by lowering workload and limiting human error. To further increase the system's practical applicability, future research can concentrate on growing the dataset, enhancing model interpretability, and implementing the system in actual clinical situations.

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